



OPEN A deep-SVM hybrid framework with enhanced EEG feature engineering and SHAP-based explainability for Alzheimer's classification

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Accurate classification among Alzheimer's Disease (AD), Fronto Temporal Dementia (FTD), and Cognitively Normal (CN) adults from EEG remains challenging. We propose a multi-class classification method that fuses interpretable spectral/connectivity biomarkers (band power, spectral entropy, α -coherence) with compact temporal embeddings from a customized lightweight one Dimensional Convolutional Neural Network (1D-CNN). The fused features are reduced by Principal Component Analysis (PCA) and classified with Support Vector Machine (SVM). All data-dependent steps like Synthetic Minority Over Sampling Technique (SMOTE), z-scoring and PCA are fit strictly on training folds to prevent leakage. Hyperparameters including PCA dimensionality and SMOTE neighbors were selected via inner-loop grid sweeps that maximized macro-F1; full grids and class distributions before/after SMOTE (inner-train only) are reported in the Supplement. On OpenNeuro ds004504 (eyes-closed; $N=88$; 36 AD/23 FTD/29 CN) the model achieved 94.5% accuracy, macro-F1 = 0.95, and AUC = 0.96. Robustness was examined on two public datasets using the identical preprocessing/segmentation: (i) cross-condition testing on ds006036 (eyes-open recordings from the same participants) and (ii) zero-shot transfer to an independent OSF dataset with different instrumentation and demographics. SHapley Additive exPlanations (SHAP) analyses provided global, class-wise, and subject-level attributions aligned with known electrophysiology (e.g., reduced α -coherence, elevated θ), supporting clinical interpretability. These results indicate that a transparent hybrid feature design can deliver accurate, leakage-safe, and explainable EEG-based differentiation of AD, FTD, and CN with preliminary external checks.

Dementia is a pressing and growing global health concern, affecting an estimated 55 million individuals worldwide, a number projected to soar to 139 million by 2050^{1,2}. Alzheimer's Disease (AD) and Fronto Temporal Dementia (FTD) are two prevalent dementia subtypes, each presenting unique clinical challenges. Early diagnosis is an important step to the process of determining the treatment plan, reducing the load on a caretaker and improving the overall quality of life of patients³. Classical methods of diagnostics like Magnetic Resonance Imaging (MRI), Positron Emission Tomography (PET) scans, or spinal fluid samples may yield useful data but are also generally expensive, invasive and are not suitable to large-scale or routine screening^{4,5}. Electroencephalogram (EEG) signals unlike other methods enables the real-time monitoring of brain signals without invasive methods. Due to its fine time resolution it is able to record initial changes in the brain regarding the development and progression of dementia^{6,7}. Such capability of the EEG in early dementia detection is illustrated in Figure 1 which highlights its applicability because it is an efficient tool with potential to offer scalable clinical neuro-diagnosis.

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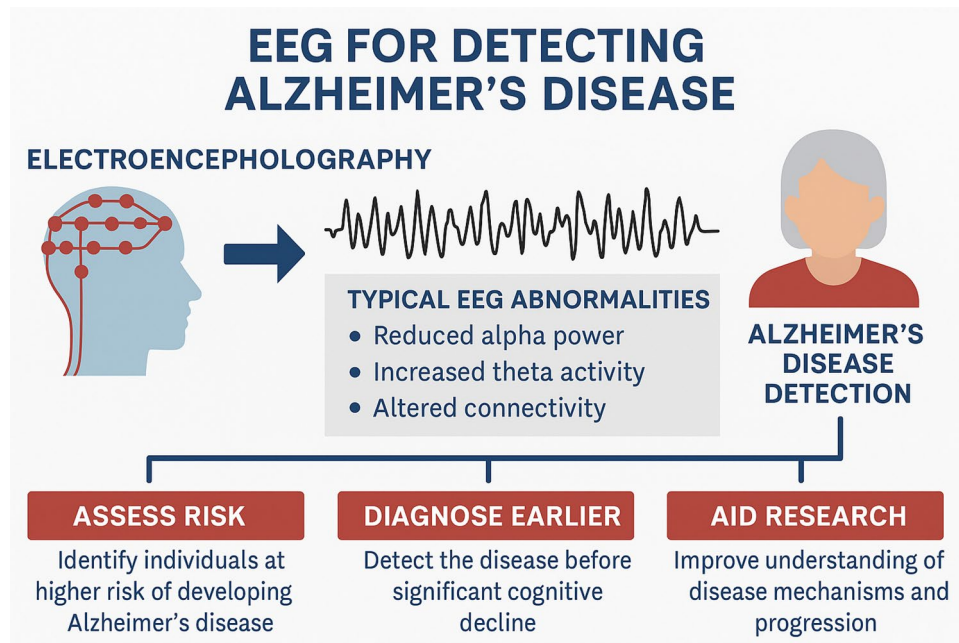


Fig. 1. Significance of EEG signals for Alzheimer's disease detection^{23–25}.

A decrease in alpha and beta waves and an increase in theta and delta power disrupt brain connectivity, which is a key indicator of dementia^{8–10}. EEG is considered a powerful diagnostic tool for the early diagnosis of dementia. But diagnosis through EEG signals remains challenging because of temporal variability, individual differences and these signals are easily affected by noise¹¹. The last stage of dementia is easy to diagnose but separating the FTD and AD is important task. Graph Convolutional Transformers are utilized by Tigga et al.¹² in order to distinguish between AD and FTD, but they utilized small and single-center dataset and reported only moderate accuracy (≈ 82 – 85%). Ghassemkhani et al.¹³ and Chetty et al.¹⁴ reported frontal–temporal asymmetry variations between AD and FTD, but their results showed limited consistency among different datasets. Current EEG based methods reported in literature have the following drawbacks (i) they mostly rely on spectral power features, (ii) they ignore the temporal aspects of neural signals, and (iii) they provide limited model interpretability. In order to mitigate the current challenges, our study suggests a hybrid Deep-SVM framework that incorporates both temporal and spectral representations into an explainable learning architecture.

For analysis of EEG signals, machine Learning (ML) based methods such as SVMs and Random Forests utilized handcrafted spectral features, producing limited improvements in accuracy^{15,16}. Deep Learning (DL) based methods like CNNs and autoencoders can learn features automatically, but they require large datasets, and due to lack of interpretability makes clinical application challenging^{17–19}. These limitations stress the need for methods that can balance interpretability and predictive performance. In this regard, hybrid frameworks that blend automated and handcrafted features are becoming increasingly popular among researchers. Recent studies have demonstrated performance gains in dementia and neurological disorder classification using such hybrid strategies^{20–22}.

Conventional machine learning methods extracted spectral power, entropy, coherence, and Hjorth parameters as features and used SVM, RF, and LDA as classifiers. These methods achieved better performance but limited generalizability across different datasets^{8,15,16}. Most works targeted AD vs. controls, with fewer directly contrasting AD and FTD, and often reported center-specific gains that did not transfer broadly. Convolutional Neural Network (CNN)/autoencoder models learn temporal–spectral representations directly from EEG, improving discrimination in some settings but typically requiring larger datasets and careful leakage control^{17,18}. For multi-class dementia, small single-center dataset frequently lead to overfitting and unstable estimates. Methods that model spatial–functional structure e.g., Graph Convolutional Networks (GCNs)/transformers over electrode graphs capture long-range dependencies and topology-aware features; however, reports on AD–FTD separation show only modest accuracy and limited replication in independent data^{12,26}. Combining handcrafted biomarkers with deep embeddings has shown consistent, incremental gains by uniting domain priors with learned dynamics, but many pipelines remain opaque and inconsistently validated^{20–22}. In this research, We target multi-class AD/FTD/CN differentiation (not only AD vs. controls). We fused interpretable spectral/connectivity biomarkers with compact 1D-CNN temporal embeddings in a single reproducible pipeline. Subject-wise, stratified nested cross-validation with all preprocessing, Synthetic Minority Oversampling Technique (SMOTE), and Principal Component Analysis (PCA) confined to the training folds has been performed. Furthermore, the proposed method provides multi-level interpretability including global, class-specific, subject-level SHapley Additive exPlanations (SHAP) that maps to known physiology. These design choices directly respond to prior limitations in generalizability, clarity, and clinical interpretability.

EXplainable Artificial Intelligence (XAI) techniques such as SHAP, Gradient-weighted Class Activation Mapping (Grad-CAM), and Layer-wise Relevance Propagation (LRP) have recently emerged as crucial bridges between algorithmic prediction and clinical reasoning²⁷. In neurodiagnostic contexts, these methods allow clinicians to visualize which EEG components most strongly influence a diagnosis—offering not just a prediction but a rationale traceable to neurophysiological phenomena. With its connection to real neurological patterns, such as alterations in alpha band coherence and deviant theta activity, XAI is able to enhance clinician trust in technology and most importantly, facilitate the interdisciplinary usage of AI solutions in everyday clinical situations. Deep architectures that are mainly developed for interpretability i.e compact xEEGNet²⁸, are a sign of the increased interest in making neuro-AI models explainable and trustworthy. Although advancement has been made, the majority of research to date still focuses on binary classification tasks (AD vs HC), rather than multi-class differentiation (AD vs FTD vs CN). Moreover, only a limited number of methods provide global, local, and temporal explanations—the three aspects recognized as key to clinical trustworthiness. The value added by this research is summarized below and offered as concrete, measurable contributions.

- This study proposed a Deep–Support Vector Machine (SVM) pipeline that extensively integrate handcrafted spectral–connectivity biomarkers with temporal embeddings attain from 1D-CNNs, achieving a substantiate accuracy of 94.5%, macro-F1 of 0.95, and AUC of 0.96 on a public multi-class dementia EEG dataset.
- The proposed method uses stratified cross-validation, PCA-based dimensionality reduction, and balanced sampling (SMOTE) to guarantee statistically sound outcomes on small-scale clinical datasets.
- In order to identify physiologically plausible EEG biomarkers (such as decreased alpha-band coherence and theta hyper-synchronization) associated with known neuropathology, global, class-specific, and subject-level SHAP studies are included.
- This study introduces interpretable EEG biomarkers that enhance diagnostic transparency, foster clinician trust, and support early diagnosis of dementia.

This study focused on the development of clinically useful diagnostic biomarkers that link the advancement of AI research with actual neurodegenerative treatment by striving for both transparency and accuracy in EEG-based dementia sub-typing.

Methodology

This study utilized a publicly available EEG dataset and proposed a robust and reliable methodology for automatically classifying subjects with FTD, AD, and CN. The pipeline integrates rigorous preprocessing, dual-path feature extraction (handcrafted and deep features), SVM classifier, and SHAP for post-hoc interpretability. Figure 2 provides an overview of the proposed architecture. For step-by-step pseudocode of training and inference, see Supplementary Methods (Algorithms S1–S2).

Dataset description

We used the publicly available OpenNeuro ds004504 v1.0.8 dataset²⁹, comprising resting-state, eyes-closed EEG recordings from 88 participants: 36 with AD, 23 with FTD, and 29 CN controls. Mini-Mental State Examination (MMSE) scores averaged 17.75 (SD = 4.5) for AD, 22.17 (SD = 8.22) for FTD, and 30 for CN subjects. The mean ages were 66.4 (SD = 7.9), 63.6 (SD = 8.2), and 67.9 (SD = 5.4) years, respectively. Median disease duration was 25 months (IQR 24–28.5). No dementia-related comorbidities were reported for AD participants. Recordings were acquired using a Nihon Kohden EEG-2100 system with 19 scalp electrodes (10–20 layout) and mastoid references A1–A2. The impedance was maintained below 5 k Ω , signals sampled at 500 Hz, with sensitivity 10 μ V/mm, time constant 0.3 s, and a high-frequency filter cutoff at 70 Hz. Recording durations averaged 13.5 min (AD), 12 min (FTD), and 13.8 min (CN), totaling 485.5, 276.5, and 402 minutes, respectively. Data were available in raw and preprocessed formats; full details are in the original dataset descriptor²⁹. Datasets' description and comparisons with other public EEG datasets are provided in Table 1. For modeling, continuous EEG was partitioned into fixed-length epochs.

We acknowledge that the per-subject counts in ds004504 (36 AD, 23 FTD, 29 CN) are modest for ML/DL. To reduce variance and guard against optimistic estimates, we adopted the following safeguards:

- classification was performed at the epoch level after strict artifact handling and 4 s segmentation, yielding 848 non-overlapping samples (303 AD, 263 FTD, 282 CN);
- all analyses used subject-wise, stratified nested cross-validation (inner tuning, outer evaluation) to prevent information leakage and to separate model selection from performance estimation;
- data-dependent steps (SMOTE, z-score normalization, PCA) were applied strictly within each training fold;
- model capacity was explicitly controlled (lightweight 1D-CNN, PCA to a compact embedding, and RBF-SVM with grid search); and
- we assessed generalization under two external regimes: cross-condition testing on the eyes-open ds006036 dataset from the same participants, and zero-shot transfer to an independent OSF dataset with different instrumentation and demographics. Together, these steps address the small-N setting by lowering estimator bias/variance, reducing leakage, and probing both within-dataset reliability and out-of-distribution robustness.

Dataset segmentation and sample generation

The OpenNeuro dataset named ds004504 (v1.0.8) includes 88 participants, of whom 36 represent the AD class, 23 belong to the FTD class, and 29 are healthy subjects. Each recording lasts 60–90 seconds and the recording

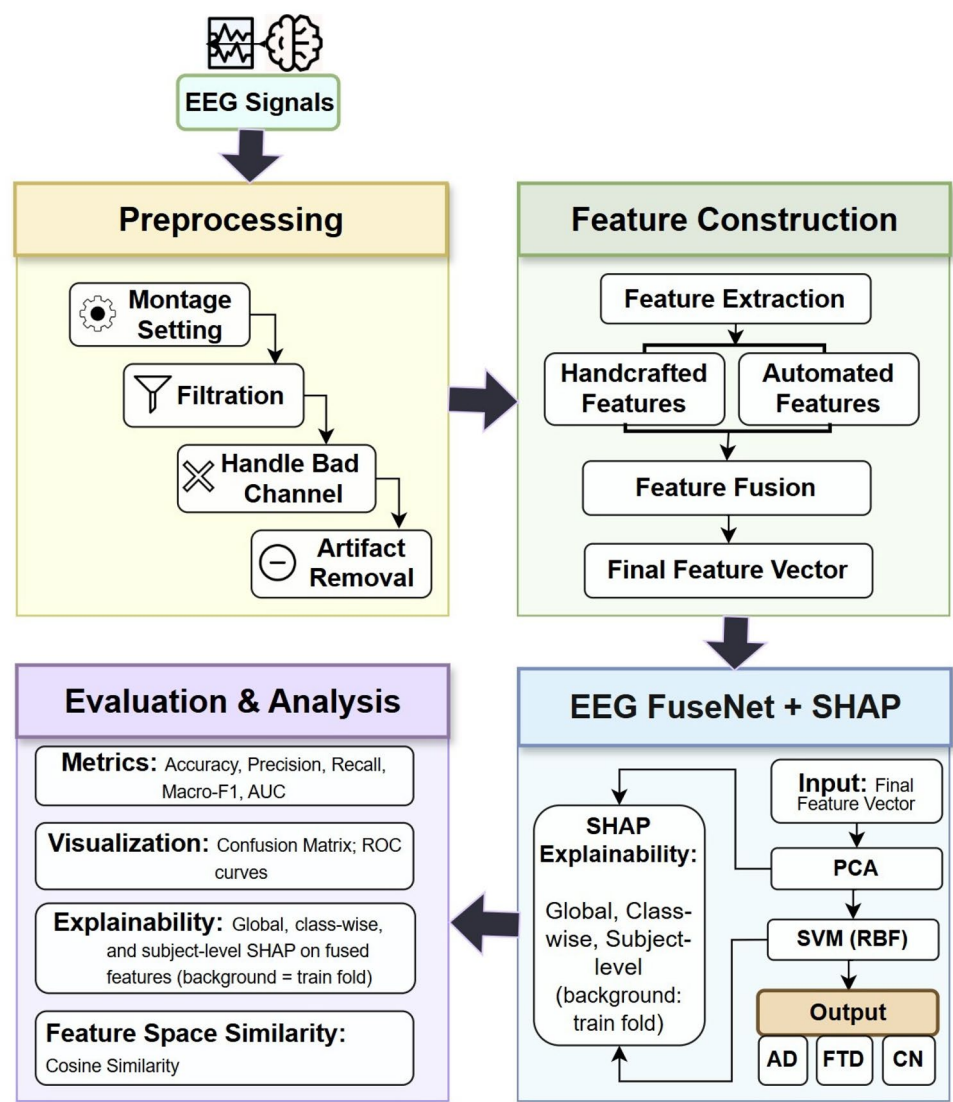


Fig. 2. Flow diagram of the proposed method.

Dataset	Classes (subjects)	Sampling rate	Duration
OpenNeuro ds004504 (Eyes-Closed) ²⁹	AD (36), FTD (23), CN (29)	500 Hz	~13 min
OpenNeuro ds006036 (Eyes-Open) ³⁰	AD (36), FTD (23), CN (29)	500 Hz	~13 min
OSF Dataset ³¹	AD (49), MCI (37), HC (23)	128 Hz	8 s
ADC ³²	AD (49), MCI (37), HC (23)	256 Hz	300 s
Cejnek ³³	AD (59), MCI (7), HC (36)	128 Hz	15 min

Table 1. Overview of publicly available EEG datasets for Alzheimer’s disease research.

was acquired during the resting state. After the removal of artifacts, EEG signals was further segmented into non-overlapping epoch of 4 s^{6,8}. And in result we got 848 clean epochs in which 303 belong to AD class, 263 from FTD class and 282 from CN class. Each epoch considered as one single observation for the extraction of relevant features. After feature fusion and the application of PCA, the final feature vector for each epoch had 128 dimensions. Classification is done at each epoch level using subject-wise nested 5-fold cross-validation. In order to prevent leakage single outer fold was considered for epoching of each subject. Fig. 6 depicts epoch-wise predictions for samples across AD/FTD/CN.

$$N_{\text{epochs}} = \left\lfloor \frac{T}{L} \right\rfloor. \tag{1}$$

where T is the post-cleaning sample count per recording.

External and cross-condition datasets

To assess the generalizability of the proposed model we have considered two additional publicly available EEG datasets were used for cross-condition and external validation. We used the eyes-open variant of the OpenNeuro dataset (ds006036) and an independent Open Science Framework (OSF) EEG dataset. To ensure the consistency of our proposed methodology both datasets followed the same preprocessing and segmentation pipeline that is described in “[Dataset description](#)”. The eyes-open ds006036 dataset includes the same 88 participants from ds004504, and these recordings were collected under same settings but with eyes open. Based on the changes in the subject vigilance and alpha activity on the brain, it is possible to evaluate the condition robustness by assessing the claim that the proposed Deep SVM model can retain the discriminative ability. Whereas the OSF dataset has an entirely different subjects and their recordings were collected with different devices and they have different demographic characteristics, which make it suitable for generalizability testing.

Each evaluation level had all the data-dependent steps restricted to the corresponding training partition. Subjects in the outer loop were divided into train/test fold (subject-wise). We performed the sequence in the inner loop (used only on the outer-train subjects): z-score fit $\rightarrow$ SMOTE (train only) $\rightarrow$ PCA fit on the inner-train set; the inner-validation split received only the transformations (z-score transform $\rightarrow$ PCA transform) and was never oversampled. The outer test fold was not touched in any way by SMOTE and was estimated using models and transforms that were fitted on the associated outer fold of training. Table 1 provides a comparison of the features of the entire large body of public EEG data pertaining to Alzheimers and dementia research with three datasets that were used in this paper underlined in bold. This is a comparative overview of where our validation strategy fits in the EEG literature at large and how each and every data set is utilized in internal and external evaluation.

EEG signals preprocessing

Raw EEG signals $x(t)$ underwent zero-phase second-order Butterworth bandpass filtering (0.5–45 Hz) to remove baseline drift and high-frequency noise:

$$H(s) = \frac{\omega_c^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2} \quad (2)$$

Implementations used a forward–backward zero-phase bandpass realized from cascaded second-order sections; the expression above denotes the canonical biquad form. The filter band (0.5–45 Hz) and referencing (A1–A2) were pre-specified on physiological grounds and not tuned. Signals were re-referenced to A1–A2 to improve spatial coherence. Artifact Subspace Reconstruction (ASR) removed transient high-amplitude artifacts^{34,35}. Independent Component Analysis (ICA) decomposed signals:

$$X = AS \quad (3)$$

where X is the EEG data matrix, A the mixing matrix, and S the independent sources. ICLabel^{36,37} automatically detected and discarded ocular and muscle artifact components, yielding clean data for feature extraction. For downstream modeling, cleaned signals were segmented into 4 s epochs. We adopted a zero-phase (forward–backward) 2nd-order Butterworth bandpass for three reasons aligned with the task and data constraints:

- (i) Physiological focus. Our analysis targets dementia-relevant bands (delta–beta); no gamma-band features are computed. Low-passing at 45 Hz excludes mains leakage (50/60 Hz) and cranial EMG that inflate high-frequency power in resting EEG, without removing task-relevant features^{8,38–40}.
- (ii) Phase integrity. Zero-phase implementation (filtfilt) eliminates phase distortion and group delay, which is critical for temporal embeddings (1D-CNN) and inter-channel measures (coherence), preserving waveform timing and relative phase.
- (iii) Stability on short epochs. A 2nd-order Butterworth provides a smooth, monotonic passband/stopband transition with minimal ringing on short 4 s segments. Higher orders (or sharp linear-phase FIR designs) require longer effective transients/taps at 500 Hz, increasing edge artifacts in short epochs; notch-only strategies suppress the line at 50/60 Hz but retain broadband EMG above ~ 45 Hz. This choice balances artifact suppression and signal fidelity for our segmentation scheme (4 s, non-overlapping), while maintaining comparability with prior AD/FTD EEG pipelines that analyze up to beta.

Handcrafted feature extraction

We extracted interpretable EEG descriptors tied to known dementia physiology. All features were computed per epoch (4 s, 500 Hz) and then concatenated into the handcrafted vector (“[Feature fusion and final feature vector formation](#)”), with z-scoring applied within the training fold. For all spectral features we used Welch’s method⁴¹ with a Hanning window of 2 s (1000 samples), 50% overlap, and $N_{\text{FFT}} = 1024$, yielding a frequency resolution of $\Delta f \approx 0.49$ Hz over the analysis band 0.5–45 Hz. Canonical bands were defined as δ (0.5–4 Hz), θ (4–8 Hz), α (8–13 Hz), and β (13–30 Hz). Low- γ was not analyzed (see “[Handcrafted feature extraction](#)”).

For channel c with one-sided PSD $P_c(f)$ (in $\mu\text{V}^2/\text{Hz}$) estimated by Welch,

$$P_c(f) = \frac{1}{L} \sum_{l=1}^L |X_{c,l}(f)|^2, \quad (4)$$

the *absolute band power* in band b is

$$\text{BP}_b(c) = \sum_{f \in b} P_c(f) \Delta f, \quad (5)$$

and the *relative band power* is

$$\text{rBP}_b(c) = \frac{\sum_{f \in b} P_c(f) \Delta f}{\sum_{f=0.5}^{45} P_c(f) \Delta f}. \quad (6)$$

We compute both BP_b and rBP_b for $b \in \{\delta, \theta, \alpha, \beta\}$ for each channel and aggregate by averaging across channels.

Spectral Entropy (SE) quantifies spectral irregularity using the normalized PSD within 0.5–45 Hz:

$$p_c(f) = \frac{P_c(f)}{\sum_{f=0.5}^{45} P_c(f)}, \quad (7)$$

$$H_s(c) = - \sum_{f=0.5}^{45} p_c(f) \log_2 p_c(f). \quad (8)$$

SE is computed per channel and averaged over channels; we also include band-limited α -SE (8–13 Hz) analogously. Hjorth parameters were computed in the time domain on band-limited epochs $x_c(t)$ (0.5–45 Hz), not in the spectral domain. For each channel,

$$\text{Activity } A = \text{var}(x_c), \quad (9)$$

$$\text{Mobility } M = \sqrt{\frac{\text{var}(\dot{x}_c)}{\text{var}(x_c)}}, \quad (10)$$

$$\text{Complexity } C = \frac{\sqrt{\frac{\text{var}(\ddot{x}_c)}{\text{var}(\dot{x}_c)}}}{M}. \quad (11)$$

We report their channel-wise means as temporal morphology indices complementary to spectral measures. Mean Alpha Coherence (MAC) is the Inter-channel functional connectivity was characterized by magnitude-squared coherence:

$$C_{xy}(f) = \frac{|S_{xy}(f)|^2}{S_{xx}(f) S_{yy}(f)}, \quad (12)$$

with auto/cross spectra estimated by Welch using the same parameters as PSD. We compute α -band coherence by averaging $C_{xy}(f)$ over $f \in [8, 13]$ Hz and then averaging across a predefined set $\mathcal{E}$ of homologous inter-hemispheric pairs $\{(F3, F4), (C3, C4), (P3, P4), (O1, O2), (T7, T8)\}$:

$$\text{MAC} = \frac{1}{|\mathcal{E}|} \sum_{(x,y) \in \mathcal{E}} \left(\frac{1}{|F_\alpha|} \sum_{f \in F_\alpha} C_{xy}(f) \right), \quad F_\alpha = [8, 13] \text{ Hz}. \quad (13)$$

This emphasizes hemispheric integration in α , implicated in AD/FTD dysconnectivity.

Deep feature learning via 1D-CNN

In parallel to handcrafted feature extraction, the filtered EEG signals were processed using a lightweight one-dimensional Convolutional Neural Network (1D-CNN) to capture temporal dependencies and discriminative frequency-domain patterns relevant to dementia-related abnormalities. The network was carefully designed to minimize overfitting given the limited size of the dataset, while still preserving a strong ability to learn and represent meaningful features.

The operation of each convolutional block can be expressed as:

$$Z^{(l)} = \text{ReLU}(W^{(l)} * Z^{(l-1)} + b^{(l)}) \quad (14)$$

where $Z^{(l)}$ denotes the output feature map of the l -th layer, $W^{(l)}$ and $b^{(l)}$ are learnable weights and biases, and ReLU is the non-linear activation function. The suggested architecture reduces feature dimensionality and allows the network to gradually incorporate temporal relationships by combining three consecutive convolutional blocks, each followed by batch normalization and max pooling layers. A dropout layer is used to reduce overfitting, and the final dense layer generates a fixed-length embedding vector that is used as input for the SVM-based classification that follows. Despite the small sample size, the network's robust feature-learning capabilities

was maintained by careful design. The proposed pipeline for deep feature extraction and classification is shown in Fig. 3. While the SVM-based training and inference procedure is shown in panel (a), panel (b) represents the 1D-CNN architecture that extracts the temporal EEG characteristics. Table 2 presents the comprehensive layer configuration, as well as filter counts, kernel sizes, and output geometry. This lightweight CNN structure allows for robust temporal feature learning and effective parameter consumption. The learned embeddings are subsequently combined with handcrafted features to enhance the discriminative capability of the final SVM classifier.

Feature fusion and final feature vector formation

To explicitly clarify the transition from handcrafted and CNN-based features to the final input representation used by the SVM, both feature sets were integrated through a reproducible normalization–concatenation–reduction pipeline. First, each handcrafted feature matrix F_H (PSD, Hjorth, SE, MAC) and CNN embedding matrix F_A was z-score normalized to ensure scale uniformity across descriptors. The two matrices were then concatenated along the feature dimension to form a joint representation:

$$F_{\text{combined}} = [F_H \mid F_A], \quad (15)$$

To remove redundancy and stabilize training, PCA was applied to F_{combined} , projecting onto a fixed 128-dimensional subspace:

$$F_{\text{final}} = \text{PCA}_{128}(F_{\text{combined}}). \quad (16)$$

This yields a compact, well-conditioned input for the subsequent SVM classifier while fusing interpretable biomarkers with data-driven temporal embeddings. Z-scoring, SMOTE, and PCA were applied strictly within training folds (inner loop), with validation/test splits receiving only the fitted transformations and no oversampling. We treated the PCA output dimensionality as an inner-loop hyperparameter and swept $k \in \{64, 96, 128, 160, 192, 256\}$ within the nested CV. The selection criterion was mean macro-F1 on inner folds with variance regularization (one-standard-error rule). $k=128$ yielded the best bias–variance trade-off:

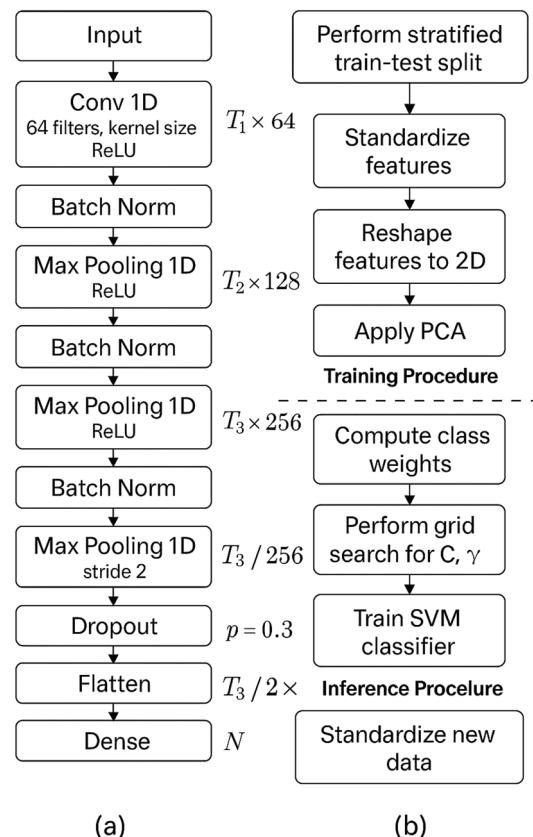


Fig. 3. Overview of the hybrid Deep-SVM feature learning and classification pipeline. **(a)** 1D-CNN architecture for EEG feature extraction, illustrating the progression from raw input to low-dimensional temporal embeddings. **(b)** SVM training and inference workflow showing standardized preprocessing, PCA-based reduction, and RBF-based classification stages. Together, these panels depict the full deep feature learning and decision process.

Layer	Filters/units	Kernel/pool	Activation	Output shape
Input	–	–	–	$(T \times C)$
Conv1D (Block 1)	64	5	ReLU	$(T_1 \times 64)$
BatchNorm	–	–	–	$(T_1 \times 64)$
MaxPool1D	–	2	–	$(T_1/2 \times 64)$
Conv1D (Block 2)	128	3	ReLU	$(T_2 \times 128)$
BatchNorm	–	–	–	$(T_2 \times 128)$
MaxPool1D	–	2	–	$(T_2/2 \times 128)$
Conv1D (Block 3)	256	3	ReLU	$(T_3 \times 256)$
BatchNorm	–	–	–	$(T_3 \times 256)$
MaxPool1D	–	2	–	$(T_3/2 \times 256)$
Dropout	–	$p = 0.3$	–	$(T_3/2 \times 256)$
Flatten	–	–	–	(N)
Dense	128	–	ReLU	(128)

Table 2. Layer-wise architecture of the 1D-CNN for EEG feature extraction (final configuration selected by inner CV).

Component	Parameter	Final (inner-selected)
1D-CNN	Conv blocks	3
	Kernels	[5, 3, 3]
	Filters per block	[64, 128, 256]
	Dropout	$p = 0.3$
	Optimizer/LR	Adam, $1e-3$; batch size 32; early stopping (patience 10)
PCA	Output dimensionality	128
SMOTE	k (neighbors)	5
	Sampling strategy	Minority up to 1.0 of majority
SVM (RBF)	C	{0.1, 1, 10} (inner-CV grid)
	γ	{0.001, 0.01, 0.1} (inner-CV grid)
	Class weights	Inverse frequency
	CV scheme	Nested subject-wise five-fold
	Objective	Maximize inner macro-F1

Table 3. Hyperparameters for the Deep-SVM hybrid (final settings reflect inner-CV selection). Full search spaces and per-fold selections are in Supplementary Table S3.

larger k offered no measurable gain and increased variance, while $k < 96$ reduced macro-F1 due to under-representation of discriminative variance. The full sweep is summarized in Supplementary Table S1.

Classification using support vector machine

Following feature extraction, classification was performed by combining handcrafted and CNN-derived features using a SVM with a Radial Basis Function (RBF) kernel. The kernel function is defined as:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (17)$$

The SVM optimization objective minimized the hinge loss:

$$\min_{w, b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i, \quad y_i(w^T \phi(x_i) + b) \geq 1 - \xi_i \quad (18)$$

Hyperparameters C and γ were optimized by grid search in the inner loop of the nested five-fold scheme. Table 3 summarizes all hyperparameters of the proposed Deep-SVM hybrid pipeline.

Cross-validation and external validation protocol

To ensure robust and unbiased performance evaluation, a nested five-fold cross-validation scheme was employed on the primary OpenNeuro ds004504 dataset. In this design, the inner loop performed hyperparameter tuning for the SVM (grid search over C and γ) while the outer loop estimated final model performance. All preprocessing and data-dependent steps—including SMOTE oversampling, z-score normalization, and PCA dimensionality reduction—were applied strictly within each training fold to prevent data leakage. To prevent data leakage, each fold was structured to keep all epochs from a single participant within either the training or testing set. Accuracy, precision, recall, AUC, and macro-F1 metrics were then assessed to average the model's performance across the outer folds. Furthermore, bootstrapping was used to determine 95% confidence intervals in order to evaluate the statistical robustness of final result.

SMOTE was only applied to the inner-train split, excluding the inner-validation and any outer-test data, and was tweaked in the inner loop using $k \in \{3, 5, 7, 9\}$. $k=5$ was chosen after much fine-tuning because it offered the best balance, producing the most consistent performance and the highest inner-fold macro-F1. Class boundaries became excessively diffused with $k \geq 7$, whereas $k=3$ tended to overfit the minority samples. The trained Deep-SVM framework was evaluated using two external evaluation setups in order to confirm the model's generalization beyond cross-validation.

- The authenticity of the model is assessed by directly evaluating it on ds006036 (eyes-open) data after it was trained on ds004504 (eyes-closed) data.
- The OSF dataset was used for a zero-shot transfer evaluation, which involved neither fine-tuning nor retraining. This test evaluated the framework's ability to generalize across various recording configurations, sampling rates, and datasets.

Figure 4 schematically illustrates the nested cross-validation and external testing workflow. This dual-level design confirms that the proposed framework not only performs reliably within the original dataset but also maintains stable classification patterns across independent EEG datasets. All data-dependent steps (SMOTE, z-scoring, PCA) were only on the training portion of each outer fold and applied to its held-out test portion; folds were subject-wise (no subject appears in both train and test). All reported performance values are averaged over the outer test folds of the nested five-fold scheme. To compare our model with the strongest reproduced baseline, we used a paired bootstrap over the outer folds of the nested CV (10,000 resamples; percentile p -value). For each resample, fold-level macro-F1 differences (ours minus baseline) were averaged and the two-sided p -value was obtained from the empirical null. The full procedure and an ablation on the number of resamples are provided in Supplementary Methods (Alg. S1), with numerical results in Supplementary Table S2.

All tunable settings were selected exclusively within the inner loop of the nested cross-validation by maximizing inner macro-F1. Domain-driven preprocessing (0.5–45 Hz zero-phase Butterworth; A1–A2 reference; ASR+ICA) was fixed a priori and not tuned. For the 1D-CNN we used a randomized grid over kernel sizes $\{3, 5, 7\}$ (per block), filters $\{[32, 64, 128], [64, 128, 256]\}$, dropout $\{0.2, 0.3, 0.5\}$, learning rate $\{1e-3, 5e-4\}$ (Adam), and batch size $\{32, 64\}$ with early stopping (patience=10) on inner macro-F1. PCA dimensionality was selected from $\{64, 128, 256\}$; SMOTE used $k \in \{3, 5, 7\}$ and class-ratio $\in [0.8, 1.0]$; SVM (C, γ) followed the inner grid in “Classification using support vector machine”. Final settings reported in the architecture and hyperparameter tables correspond to the *inner-selected* configuration most frequently chosen across folds (per-fold selections are summarized in Supplementary Table S3). All data-dependent steps (SMOTE, z-scoring, and PCA) were fit only on the training portion of each outer fold and then applied to the corresponding held-out test split. We report accuracy, macro-F1, macro-precision, macro-recall, and one-vs-rest macro-AUC. Macro-averaging mitigates class-frequency effects and weights AD/FTD/CN equally.

Explainability with SHAP

To improve clinical trust, SHAP values⁴² were computed to attribute predictions to individual features:

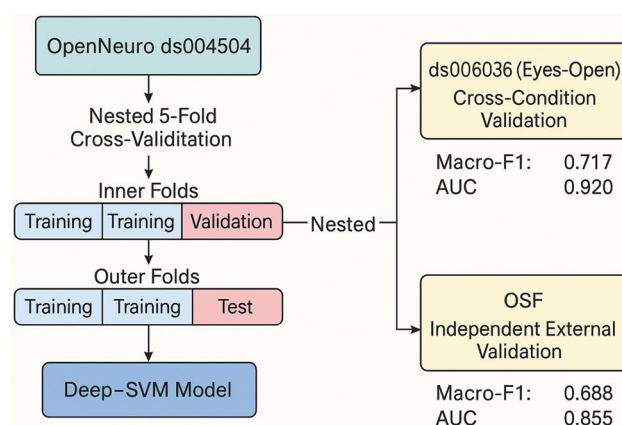


Fig. 4. Illustration of the nested cross-validation (inner/outer folds) and external validation pipeline.

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (19)$$

This provided both global (dataset-level) and local (subject-level) interpretability, allowing neurologists to trace model decisions back to physiologically meaningful EEG patterns. For reporting, per-epoch SHAP values are (i) grouped by physiologically meaningful units (band $\{\delta, \theta, \alpha, \beta\} \times$ region: frontal, central, parietal, occipital, temporal; and connectivity pairs for α -coherence), (ii) averaged across an epoch set for the subject, and (iii) sign-preserved to indicate risk direction (positive $\rightarrow$ pushes toward the predicted class, negative $\rightarrow$ away). The clinician-facing report comprises three panels: (A) calibrated class probabilities (AD/FTD/CN); (B) a top-10 feature waterfall plot with band/region labels; and (C) a scalp map showing aggregated SHAP by band (e.g., reduced posterior α power/coherence or increased frontal θ). This design mirrors routine EEG review (band-specific, region-wise) and can be rendered as a one-page PDF (see Supplementary Fig. S1). Clinician-facing outputs include per-subject class probabilities and SHAP waterfall plots for the top-10 features, with scalp maps of band-specific SHAP attributions (see Supplementary Fig. S1). Overall, the proposed framework combines robust preprocessing, complementary handcrafted and deep features, and explainable classification, providing a reliable and interpretable tool for EEG-based dementia subtype detection.

Results and discussion

The experiments were conducted on the publicly available OpenNeuro dataset²⁹, consisting of resting-state, eyes-closed EEG recordings from 88 participants: 36 diagnosed with AD, 23 with FTD, and 29 CN controls. Each recording was obtained using 19 scalp electrodes arranged according to the 10–20 system, sampled at 500 Hz, with session durations averaging 12–14 min per subject. This dataset offers clinically validated diagnostic labels and high-quality neurophysiological data, enabling a robust evaluation of the proposed framework (Figs. 5, 6, 7, 8).

When tested on this dataset, the suggested Deep-SVM hybrid model effectively and reliably distinguished between AD, FTD, and CN individuals. First, how signal preprocessing affected the quality of EEG data was investigated. Bandpass filtering and ICA-based artifact reduction were used to preserve physiologically significant brain oscillations while successfully suppressing noise and ocular artifacts. By lowering the possibility of mistakes brought on by lingering artifacts, these preprocessing techniques produced cleaner EEG data for feature extraction and classification. Every component in the suggested pipeline contributes differently, as seen by the ablation analysis in Table 4. Performance decreased when CNN-derived or handcrafted features were utilized separately, indicating that combining data-driven and domain-specific representations offers complementing discriminatory power. Due to class imbalance, removing SMOTE decreased performance, and leaving out PCA resulted in redundancy and poorer AUC values. The best overall performance was obtained by the full configuration, which combined CNN and handmade features, balanced sampling, and PCA before SVM classification. It consistently outperformed all other setups, with an accuracy of 94.5%, a macro-F1 of 0.95, and an AUC of 0.96.

According to the leakage control measures, Without SMOTE configuration still yielded competitive results, suggesting that oversampling artifacts were not the main cause of the observed performance benefits. To further confirm robustness, inner-loop experiments varying PCA dimensionality and SMOTE neighborhood parameters showed that the chosen configuration (PCA $k=128$, SMOTE $k=5$) consistently delivered the best inner-CV macro-F1 score with low variance. Detailed results of these sweeps are provided in Supplementary Table S1. The results in Table 5 demonstrate that the proposed In nested 5-fold cross-validation (outer test folds), the Deep-SVM hybrid framework continuously surpassed all other models. With the lowest standard deviation, the model showed reliable excellent performance, obtaining an outstanding mean accuracy (94.5%), macro F1-score (0.95), and AUC (0.96). Supplementary Table S2 indicates a statistically significant enhancement (percentile $p < 0.05$) between the suggested model and the best-performing replicating baseline (macro-F1),

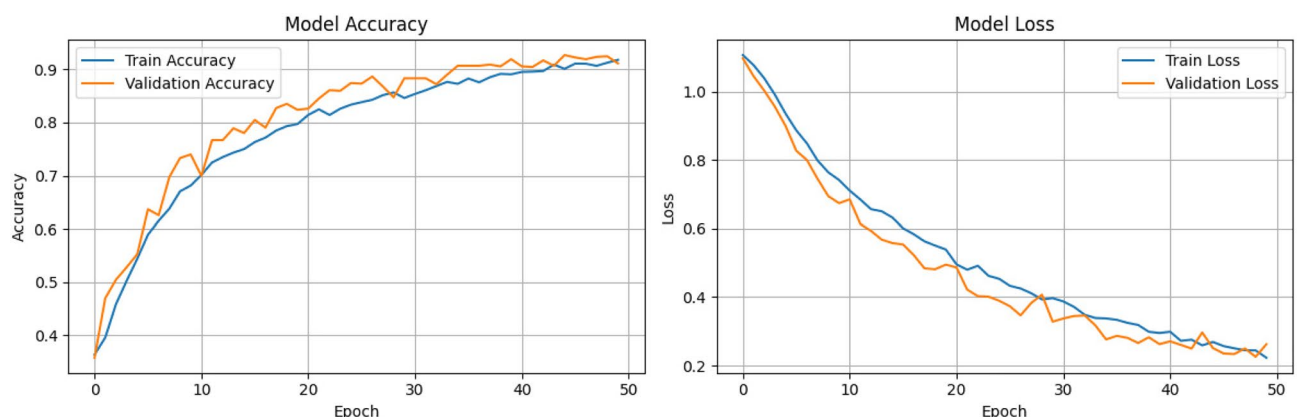


Fig. 5. Training and validation accuracy trends of the Deep-SVM hybrid model over multiple epochs.

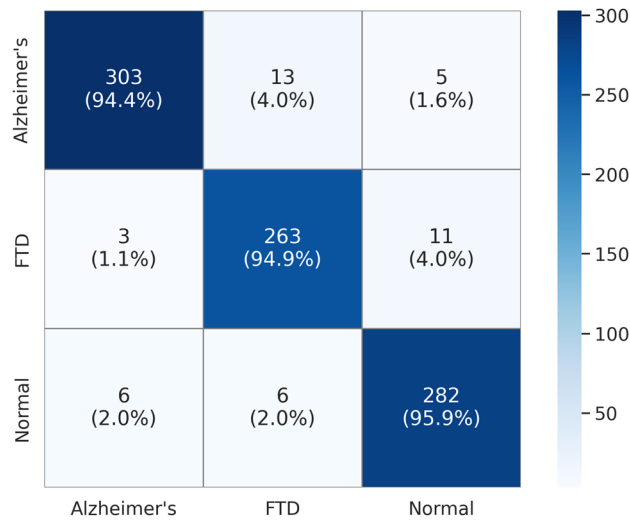


Fig. 6. Confusion matrix illustrating how the model correctly and incorrectly classified samples among the AD, FTD, and CN groups.

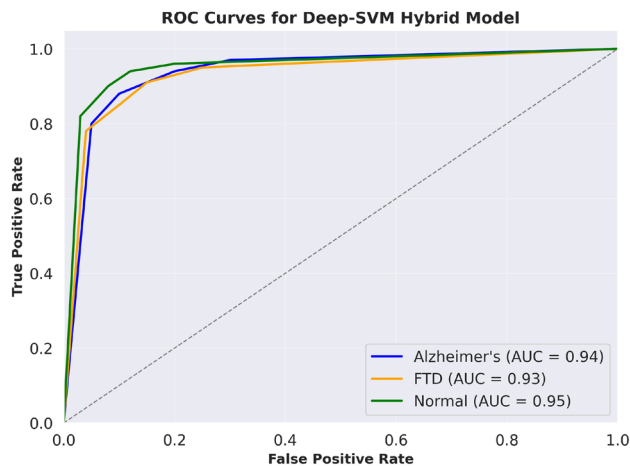


Fig. 7. ROC curves illustrating the trade-off between sensitivity and specificity for each class.

as identified via a paired bootstrap analysis. An important requirement for clinical deployment, these results offer compelling proof that the suggested approach produces consistent and trustworthy results in addition to improved accuracy. Every stated measure is an average calculated throughout the nested 5-fold cross-validation scheme's outer test folds (Table 6).

Even with a relatively small sample size, the hybrid model appears to generalize well to unseen data, as evidenced by the training and validation curves in Fig. 5, which show stable convergence without indications of overfitting. It is worth noting that the internal validation accuracy in Fig. 5 is slightly higher than the training accuracy. This pattern is expected in our setting due to regularization (dropout) and early stopping in the 1D-CNN encoder, together with strict nested subject-wise cross-validation. Moreover, SMOTE is applied only within training partitions, which can make the effective training distribution more challenging while improving generalization on held-out folds. Results from cross-condition evaluation on OpenNeuro ds006036 (eyes-open) and independent external testing on the OSF EEG dataset are reported separately in the sections "External and Cross-Condition Datasets" and "Cross-Validation and External Validation Protocol" and are not included in Fig. 5.

A thorough summary of the SHAP feature relevance among the groups with AD, FTD, and CN is given in Fig. 9. The primary EEG principal components (PCs) that influence the Deep-SVM model's predictions within each diagnostic category can be easily compared in Panels (a)–(c). Figure 10 displays the decision trajectories that illustrate how individual PCs gradually push the model's output in the direction of or away from a specific diagnostic conclusion. While the CN group showed more stable trajectories with fewer dominant factors, only a small number of PCA-derived components had a significant impact on the final decision in both AD and FTD. By identifying the EEG elements that most affect the model's diagnostic reasoning, these findings collectively provide unambiguous, case-level interpretability.

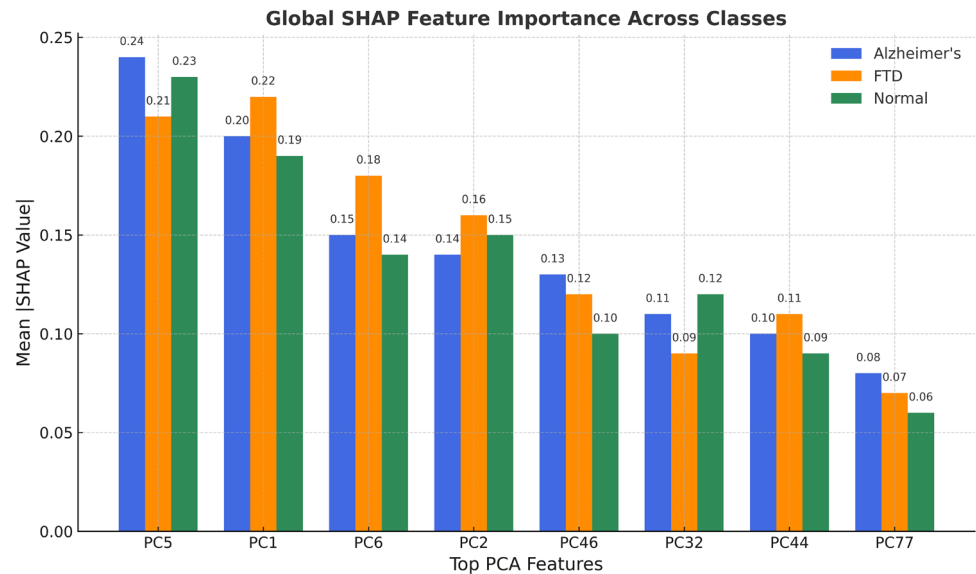


Fig. 8. Global SHAP feature importance plot showing the key EEG components that most strongly influence the model’s classification decisions.

Experiment configuration	Accuracy (%)	Macro-F1	AUC
Handcrafted features only + SVM classifier	90.50	0.89	0.91
CNN features only + DNN classifier	91.20	0.90	0.92
Without SMOTE (imbalanced data)	91.00	0.89	0.90
Without PCA (no dimensionality reduction)	92.30	0.91	0.94
Handcrafted + CNN features without fusion	91.50	0.90	0.92
Bandpass + handcrafted features only + SVM	87.80	0.85	0.88
Bandpass + artifact removal + CNN features only	89.10	0.86	0.89
Feature engineering only + ensemble classifier	92.60	0.92	0.94
Feature selection using PCC only + SVM classifier	92.70	0.92	0.93
Feature selection using PCC + PSO + SVM classifier	93.40	0.94	0.94
Feature selection using PCC + PSO + CNN classifier	93.80	0.94	0.95
SMOTE + feature engineering only (no fusion) + SVM	90.90	0.88	0.90
CNN-based feature extraction + PCA + DNN classifier	91.90	0.90	0.93
Handcrafted + CNN + SMOTE (w/o PCA) + LSTM classifier	92.60	0.92	0.94
Proposed: Handcrafted + CNN features + SMOTE + PCA + Deep-SVM hybrid	94.50	0.95	0.96

Table 4. Ablation study across preprocessing, feature selection, fusion, and classifier choices. Note (PCC and PSO). PCC ranks features by absolute Pearson correlation with one-vs-rest class indicators; highly collinear pairs (e.g., $|r| > 0.90$) are pruned. PSO is a wrapper: each particle encodes a binary feature mask; the objective maximizes 5-fold CV macro-F1 with a sparsity penalty; velocities update as $v^{t+1} = \omega v^t + c_1 r_1 (pbest - x^t) + c_2 r_2 (gbest - x^t)$ and masks are binarized via a sigmoid threshold.

We first determined which spatial EEG components had the most influence on model predictions, and then we looked at the times of the EEG recordings when these discriminative patterns were most active. For typical individuals from each diagnostic group, Fig. 11 illustrates the evolution of SHAP values over time. In the early and mid-latency windows, in particular, distinct temporal patterns with higher SHAP contributions emerged in the Alzheimer’s and FTD groups, indicating stronger disease-related brain responses than those seen in controls. PC5, PC6, and PC14, the PCA components that were most influential during these times, corresponded to disturbances in frontal-temporal coherence and alpha and theta oscillations that are frequently associated with neurodegeneration. These results suggest that the Deep-SVM framework may be able to identify the emergence of abnormal neural synchronization, providing biologically meaningful temporal biomarkers for distinguishing between stages of progression.

An extensive multi-panel graphic was created by combining all subject-level SHAP waterfall renderings (Fig. 12). Representative cases for participants with AD, FTD, and CN are shown in panels (a)–(c). Each panel displays the contribution of particular EEG Principal Components (PCs) to the final result, offering a

Model configuration	Accuracy (%)	Macro F1	AUC	p-value (bootstrap)
Handcrafted + SVM	90.5 ± 1.2	0.89 ± 0.02	0.91 ± 0.01	< 0.001
CNN Features + DNN	91.2 ± 1.0	0.90 ± 0.03	0.92 ± 0.02	< 0.001
CNN + PCA + DNN	91.9 ± 0.8	0.90 ± 0.02	0.93 ± 0.02	0.002
Handcrafted + CNN + SMOTE + LSTM	92.6 ± 0.7	0.92 ± 0.02	0.94 ± 0.01	0.015
Proposed model (deep-SVM hybrid)	94.5 ± 0.5	0.95 ± 0.01	0.96 ± 0.01	–

Table 5. Performance comparison with statistical significance analysis was carried out across multiple runs (nested five-fold cross-validation; outer test folds). The results’ Standard Deviation (SD) is shown as mean ±. In order to compare the outer folds (10,000 resamples; percentile method) with the suggested model, the *p*-values were obtained using a paired bootstrap procedure.

Class	Precision	Recall	F1-Score	Support
Alzheimer’s	0.94	0.95	0.94	303
FTD	0.96	0.95	0.95	263
Normal	0.95	0.95	0.95	282
Accuracy	0.95			848
Macro Avg	0.95	0.95	0.95	848
Weighted Avg	0.95	0.95	0.95	848

Table 6. Classification report of the Deep-SVM model on the EEG dataset. High precision and recall across all classes clearly show that the model classifies each category effectively and consistently.

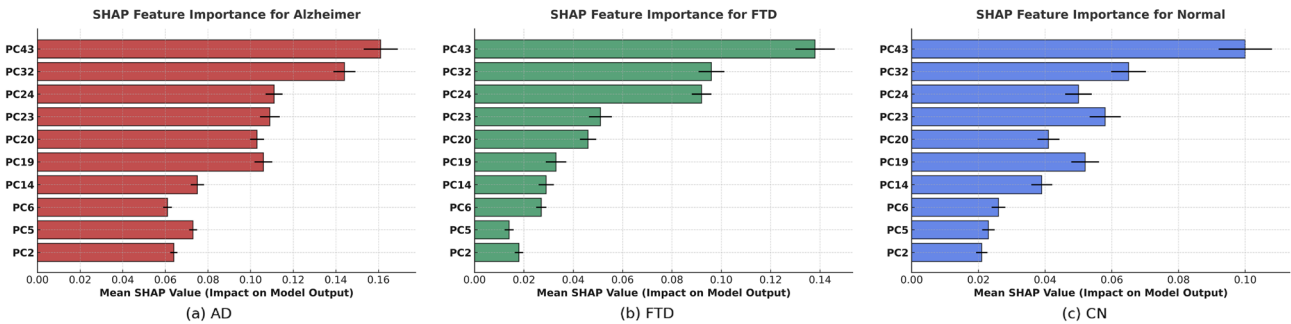


Fig. 9. Consolidated SHAP feature importance across classes. (a) AD, (b) FTD, (c) CN. The unified layout facilitates cross-class comparison of dominant EEG principal components contributing to the Deep-SVM predictions.

unique perspective of how the Deep-SVM model arrived at its prediction. In contrast to negative contributions (blue bars), which have the opposite impact, positive contributions (red bars) boost the model’s confidence in allocating a subject to a particular class.

For participants with AD and FTD, components linked to theta hyper-synchronization and disrupted frontal-temporal coherence produced strong positive influences that drove class-specific predictions. In contrast, CN participants exhibited more balanced contributions, resulting in outputs that remained closer to the baseline expectation. Collectively, these visualizations enhance the clinical interpretability of the Deep-SVM framework by revealing which EEG features play the most influential role in guiding individual diagnostic decisions.

The identified EEG patterns and their SHAP attributions align with widely recognised dementia biomarkers: reduced posterior α power/coherence and increased frontal θ have been repeatedly linked to AD/FTD pathophysiology. We operationalise these findings into a clinician-facing report that includes: (i) calibrated class probabilities (assistive, not definitive); (ii) a top-features waterfall plot where each bar carries a band $\times$ region label (e.g., “ θ power: frontal”); and (iii) a scalp map highlighting aggregated SHAP contributions per band. To aid actionability, we summarize interpretation rules in Table 7: SHAP sign indicates whether a feature increases the likelihood of a class (positive) or is protective (negative); band/region context provides clinical meaning (e.g., posterior α $\downarrow$ or frontal θ $\uparrow$). In practice, this one-page output can be reviewed alongside conventional EEG traces and spectral summaries. We emphasize that the tool is *decision support*: it complements but does not replace clinical judgment, and it flags patterns for targeted review rather than issuing diagnostic labels in isolation.

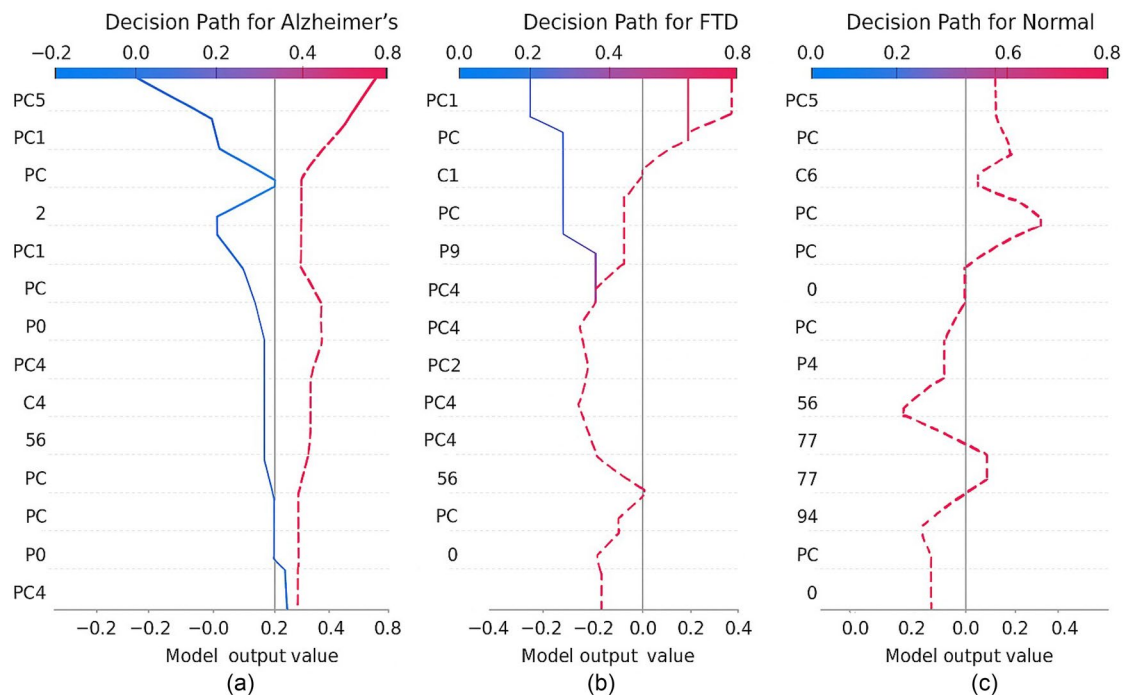


Fig. 10. The Deep-SVM model's local decision path plots for (a) AD, (b) FTD, and (c) CN predictions display the cumulative contributions of the main EEG-derived principal components. Positive contributions increase the likelihood of the prediction moving toward the target class, whereas negative contributions decrease it.

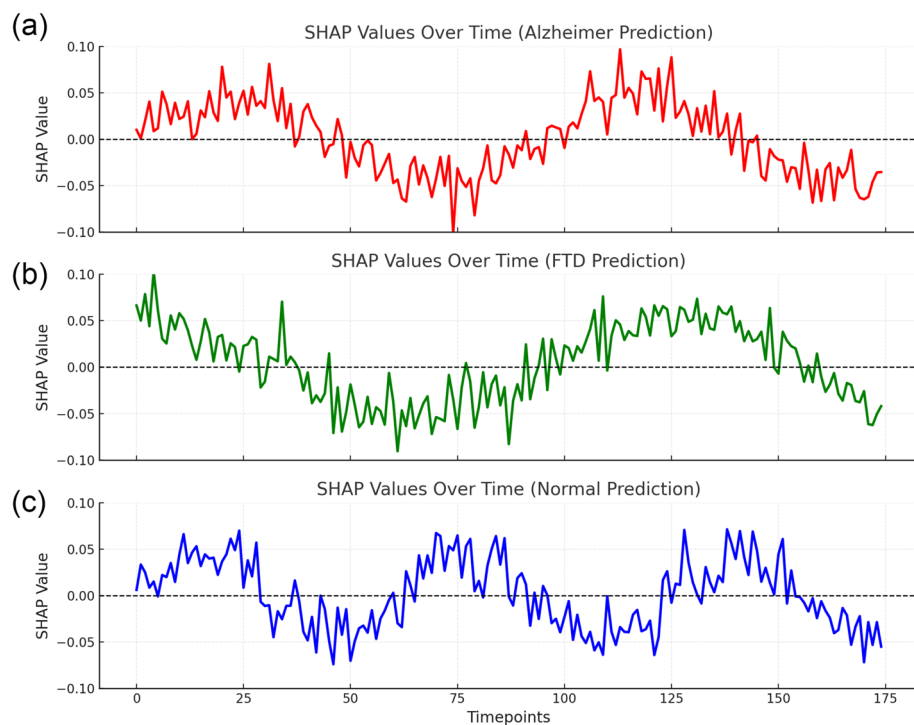


Fig. 11. SHAP value distributions over time for different prediction classes: (a) FTD prediction, (b) Normal prediction, and (c) Alzheimer's prediction. Positive SHAP values represent timepoints that contribute more strongly to the predicted class, whereas negative values reflect timepoints that reduce the likelihood of that prediction.

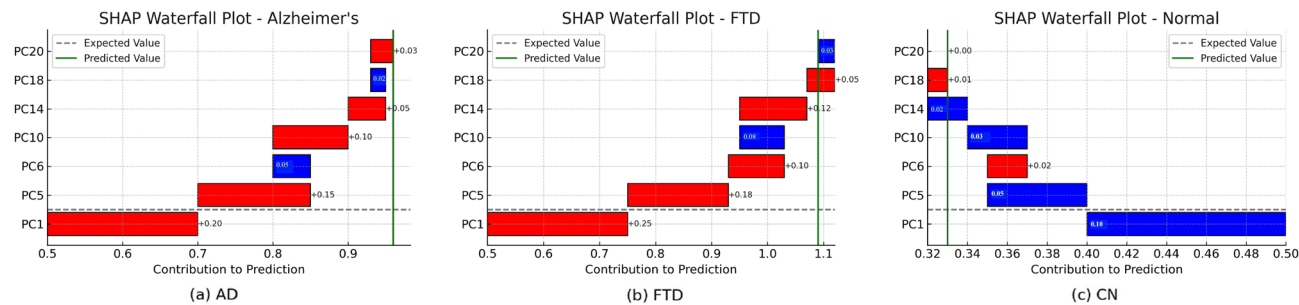


Fig. 12. SHAP waterfall plots that have been combined for representative themes. AD, FTD, and CN are the three variables that make up (a). Positive and negative contributions to the predicted value are indicated by red and blue bars, respectively.

Feature (band × region)	SHAP sign	Typical EEG finding	Clinical note
θ power: frontal	+ (AD/FTD)	Elevated slow activity	Consistent with diffuse slowing; review meds/sleep; consider AD/FTD risk if persistent
α power: occipital/parietal	– (AD/FTD)	Reduced posterior α	Loss of posterior dominant rhythm strength; aligns with cortical dysfunction
α coherence: inter-hemispheric pairs (P3–P4, O1–O2)	– (AD/FTD)	Reduced α connectivity	Dysconnectivity marker; supportive when combined with posterior α reduction
β power: central	mixed	Variable beta modulation	Interpret with meds/EMG context; low weight relative to α/θ
θ/α ratio: frontal/posterior (derived)	+ (AD/FTD)	Increased slow-to-alpha ratio	Robust composite indicator of cognitive impairment risk

Table 7. Clinician-facing summary linking SHAP patterns to EEG findings and practical notes. SHAP sign is with respect to the predicted class.

Study	Task	Subjects	Protocol	Acc. (%)	F1	Method
Zhang et al. (2023) ²⁰	Binary	100	Record-wise 10-fold	91.5	NR	1–45 Hz; STFT; CNN; <i>record-wise split (optimistic vs. subject-wise)</i>
Kumar et al. (2024) ²³	AD vs HC	88	Subject-wise CV	78.3	NR	0.5–40 Hz; ICA; classical ML
Puri et al. (2023) ⁴³	AD vs HC	88	Holdout	90.8	NR	0.5–45 Hz; PSD/coherence; CNN+FC
Ali et al. (2024) ⁴⁴	Multi-class	52	Subject-wise CV	88.7	0.92	Band powers; deep classifier
Jiao et al. (2023) ³⁹	Multi-class	109	Subject-wise CV	89.9	NR	Sensitivity reported macro; DL
Stefanou et al. (2025) ⁴⁵	Multi-class	88	Subject-wise CV	80.0	0.68	Classical features; ML
Miltiadous et al. (2023) ⁴⁶	Multi-class	88	Subject-wise CV	83.5	0.85	Classical features; ML
Chen et al. (2023) ⁴⁷	Multi-class	88	Holdout	76.5	NR	Sensitivity macro; DL
Lal et al. (2023) ⁴⁸	Multi-class	102	Subject-wise CV	87.9	0.91	Classical+DL hybrid
Hemant et al. (2024) ⁴⁹	Multi-class	69	Subject-wise CV	89.1	0.90	DL
Proposed (Deep-SVM)	3-class (AD/FTD/CN)	88	Nested subject-wise five-fold	94.5	0.95	0.5–45 Hz; ASR+ICA; PSD/Hjorth/SE/ α -coh + 1D-CNN; PCA(128)+SVM

Table 8. Comparison with prior EEG-based AD/FTD studies. We list task, dataset size N subjects, validation protocol, and reported metrics to clarify non-comparable design choices across works. NR, not reported; “(window-wise)”, paper reports only window/epoch counts.

Table 8 benchmarks the proposed approach against existing state-of-the-art methods. While prior EEG-based classifiers reported accuracies ranging from 76 to 91%, our hybrid model reached 94.5% accuracy with improved F1-score and sensitivity, particularly benefiting the minority FTD class. This performance gain is attributed to synergistic feature fusion, class balancing, and the SVM’s capacity to define optimal decision boundaries in high-dimensional spaces. Direct comparison with prior EEG dementia studies is nontrivial because of heterogeneity in (i) datasets and cohort sizes (single- vs. multi-center; class composition), (ii) preprocessing choices (filters, re-referencing, artifact removal), (iii) feature spaces (PSD/connectivity vs. end-to-end deep embeddings), and (iv) evaluation protocols (subject-wise vs. epoch-wise splits; k -fold vs. LOSO; macro-F1 vs. accuracy/AUC). To aid readers, Table 8 enumerates these axes for frequently cited baselines we reproduced or referenced. Our comparisons therefore emphasize *within-dataset*, *subject-wise nested CV* under a leakage-safe pipeline and report macro-F1 alongside accuracy and AUC to accommodate class imbalance, while using the same preprocessing and segmentation for all reproduced baselines.

To quantify differences, we performed a paired bootstrap test over outer folds (10,000 resamples) comparing our macro-F1 to the strongest reproduced baseline; the improvement was significant ($p < 0.05$; see Supplementary Table S2). Table 8 benchmarks the proposed approach against existing EEG methods. While prior classifiers reported 76–91% accuracy under heterogeneous protocols, our hybrid model reached 94.5% accuracy with macro-F1 = 0.95 and sensitivity = 0.96, with the largest gains on the minority FTD class. We attribute this to synergistic feature fusion, within-fold class balancing, and the RBF-SVM's margin in a compact PCA space.

Overall, these findings demonstrate that combining handcrafted and CNN-derived features, balanced through SMOTE, dimensionally reduced by PCA, and classified with an SVM, results in a highly accurate and interpretable EEG-based dementia diagnostic framework. Through the integration of many SHAP analysis layers, such as global, local, temporal, and subject-specific visualizations, the framework provides profound understanding of the decision-making process and adds a remarkable degree of transparency to the model's reasoning process. The suggested Deep-SVM framework is positioned as a therapeutically promising and explicable diagnostic tool for neurodegenerative illnesses due to its multi-layered interpretability and outstanding predictive performance. Future work should incorporate statistical significance testing on larger, multi-center datasets to further validate these results and strengthen claims of generalizability.

Conclusion and future work

We introduced an interpretable EEG pipeline that combines domain-informed spectral/connectivity biomarkers with lightweight 1D-CNN temporal embeddings, followed by PCA compression and RBF-SVM classification. Using subject-wise, stratified nested cross-validation with all preprocessing confined to training folds, the approach reached 94.5% accuracy (macro-F1 = 0.95, AUC = 0.96) on OpenNeuro ds004504. Beyond internal validation, we examined robustness in two public datasets: (i) cross-condition testing on ds006036 (eyes-open) and (ii) zero-shot transfer to an independent OSF dataset differing in hardware and demographics. SHAP-based attribution consistently highlighted physiologically plausible patterns (e.g., α -band coherence reduction and θ alterations), strengthening clinical interpretability.

These findings suggest that hybrid, leakage-safe EEG modeling can support reliable three-way differentiation among AD, FTD, and CN while providing transparent, clinician-readable rationales. Future directions include multi-center expansion to broaden demographic and hardware diversity, longitudinal analyses for progression tracking, and multi-modal fusion (EEG with MRI or fluid biomarkers). Translational work will focus on streamlined, real-time dashboards that surface per-patient SHAP summaries and uncertainty estimates to facilitate adoption in memory clinics.

Data availability

The datasets generated and/or analyzed during this study are available in the repository at <https://openneuro.org/datasets/ds004504/versions/1.0.8>, https://openneuro.org/datasets/ds006036/versions/1.0.2?utm_, https://osf.io/2v5md/?utm_.

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Author contributions

F.A. conceived the study, performed data preprocessing, implemented the experimental framework, and drafted the manuscript. Y.A. contributed to methodology refinement, statistical analysis, and interpretation of results. S.M.U. supervised the research, guided experimental design, and contributed to manuscript structuring. S.K. provided technical guidance, validated the methodology, and oversaw project administration. I.I. assisted in data curation, feature extraction, and performance evaluation. M.A.A. contributed to result interpretation and visualization. All authors contributed to writing, reviewing, and editing the manuscript, and approved the final version.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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